SOMAIYA VIDYAVIHAR UNIVERSITY

15·05·2025 (E)

Semester: January 2025– April 2025			
Maximum Marks: 50	Examination: End-Semester Examination		Duration: 2 Hrs.
Programme code: H02 Programme: Honours in CSF	Class: SY		Semester: IV (SYU-2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department/Section/ Center: COMP/CCE		
Course Code: 216H02C401	Name of the Course: Applied Cryptography		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Differentiate between substitution cipher and transposition ciphers. Support your answer with examples of each.	05
ii)	Discuss significance of following algorithms in applied cryptography; also state at least one example algorithm from the mentioned family. a. Primality testing b. Prime factorization.	05
iii)	Explain the following security requirements of a cryptographic hash function: i) Pre-image resistance ii) Second pre-image resistance iii) Collision resistance	05
Q.2	Attempt any **one**	10
i)	Consider the scenario: A professor encrypts a lecture video, uses a simple AES encryption tool to lock the file with a shared password known only to her class. The university has a cloud storage system, which is backed by a reliable content delivery network (CDN). This ensures that students, whether in the hostel, at home, or on a slow connection, can access the video whenever they need it. No user login is required to download the video. The university doesn't track who watches it. Any student with the decryption password can play the lecture on their own device using a basic player that decrypts the file on the fly. Its ok if a few frames of the video get corrupted during download, the lecture plays through. A. Comment on following security goals if they are achieved or not. Justify your answer in 1-2 statements. a. Confidentiality b. Integrity c. Availability d. Authentication e. Non-repudiation of sender f. Non repudiation of receiver	06
	B. For every goal not achieved in above list, <u>list</u> two mechanism those might achieve them. [No explanation is expected]	04

ii)	Explain the AES process in brief. Discuss any two real life applications wherein AES could be a good choice to ensure security of data.	10
Q.3	Attempt any **one**	10
i)	Describe the structure and purpose of a digital certificate. What is the role of a Certificate Revocation List (CRL) and why is it important?	5+5
ii)	Explain the Kerberos authentication process involving the Key Distribution Center (KDC). Include roles of TGT and service tickets. Support your answer with a neat diagram.	10
Q.4	Attempt the following	10
	A. Consider a RSA algorithm wherein the value of n is 323. Use RSA algorithmic concepts to choose a public key 'e'.	5
	B. For an ECC curve defined over Zp with the equation- $y^2 = x^3 + 4x + 4$ mod 5; Compute all points those might form the curve.	5
Q.5	Attempt the following	10
	A. Define Homomorphic Encryption. Describe a scenario where Homomorphic Encryption can be particularly useful.	05
	B. What is Secure Multi-Party Computation (MPC)? Provide an example where MPC could be used to enable privacy-preserving computations.	05